

BLOCKCHAIN / WEB3

Verification Assumes Settled Residency

A portable, continuity-based identity layer for the transnational workforce funding cross-border corridors.

Team : Error 404

Google Hackathon 2026 · Track: Blockchain / Web3

The Problem

Identity verification assumes a person has one settled legal home.

- A fixed address.
- A national ID anchored to that address.
- A bank account tied to one jurisdiction.

But the workforce funding remittance and gig corridors is transient by nature — visas expire, permits change, jobs move across borders.

The Mismatch

**Verification checks
a single snapshot
of residency.**

*These workers need continuity verified
across jurisdictions that keep changing.*

Who Hits This Wall

1

Gig & Platform Worker

Earns across three countries in a year via delivery and freelance apps. No single address to anchor a KYC check.

2

Seasonal Migrant Labor

Work permits renew every 6-9 months across different states. Bank onboarding resets each time.

3

Remittance Sender

Sends money home monthly but is repeatedly re-verified because prior proofs 'expired' with their visa.

Verify continuity, not residency.

Instead of asking “where do you legally reside right now?”, ask “can we prove this is the same accountable person across every jurisdiction they've touched?”

This turns identity from a single brittle claim into a portable, resilient chain of attestations — exactly what blockchain is built to anchor.

The Solution

Portable Continuity Passport

01

Identity Layer

Self-sovereign DID — the worker owns their identity wallet, not tied to any one state.

02

Attestation Layer

Verifiable Credentials from any issuer: employer, bank, embassy, NGO, cooperative.

03

Continuity Engine

On-chain scoring of gaps, overlaps, and revocations across the credential trail.

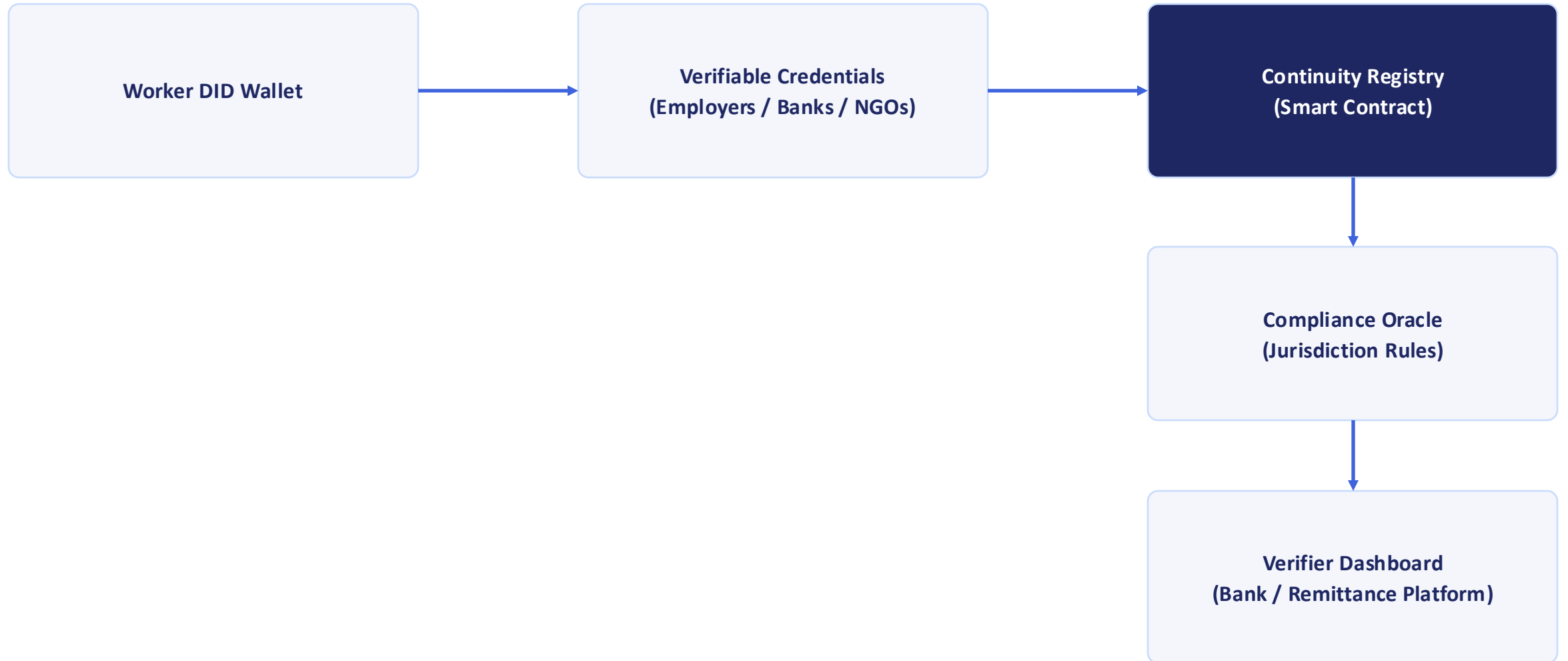
04

Compliance Oracle

Translates a destination jurisdiction's real legal rule into an on-chain checkable condition.

A fifth layer, the Verifier Interface, queries sufficiency of proof without seeing the worker's full history.

Architecture



Flow: the worker's wallet accumulates credentials → anchored in the Continuity Registry → checked against jurisdiction rules → verifier gets a sufficiency proof.

How It Works



Why Blockchain



No single jurisdiction owns the record

Critical since the worker's problem is that no one state claims them permanently.



Auditable issuance & revocation

No central authority has to arbitrate disputes between countries.



Zero-knowledge privacy

Prove sufficiency of history without exposing every employer or visa detail.



True portability

The identity wallet moves with the person, not with a bank account that gets frozen on exit.

Compliance Positioning

We are not building a KYC/AML bypass.

Complementary, not a workaround

Designed to sit alongside FATF Travel Rule compliance, feeding it better data — not routing around it.

Risk-based tiering, same as regulators already use

Low-value transactions need thin continuity proof; high-value or cross-border corridors need thicker proof.

Sanctions & AML checks stay separate

This system proves continuity of identity. It does not replace screening — that remains a distinct, layered check.

Demo Scope

Built for this hackathon vs. the production vision

Working Today

- DID creation + credential issuance flow (testnet)
- Continuity Registry smart contract, deployed
- Verifier dashboard with sufficiency checks
- 3 mocked jurisdiction rule-sets (JSON)

Simulated / Roadmap

- Zero-knowledge proof circuit (mocked interface for demo)
- Live compliance oracle feeds (real regulatory APIs)
- Production biometric onboarding
- Real issuer partnerships (banks, embassies, NGOs)

Roadmap & Ask

Next 3 months

Harden the ZK continuity circuit; pilot with one remittance corridor.

6 months

Onboard 2-3 real issuers (employer, cooperative, NGO) in a single corridor.

12 months

Multi-corridor rollout; work with a regulator sandbox on the compliance oracle.

We're looking for: pilot corridor partners, ZK/cryptography mentorship, and regulatory feedback.

Thank you.